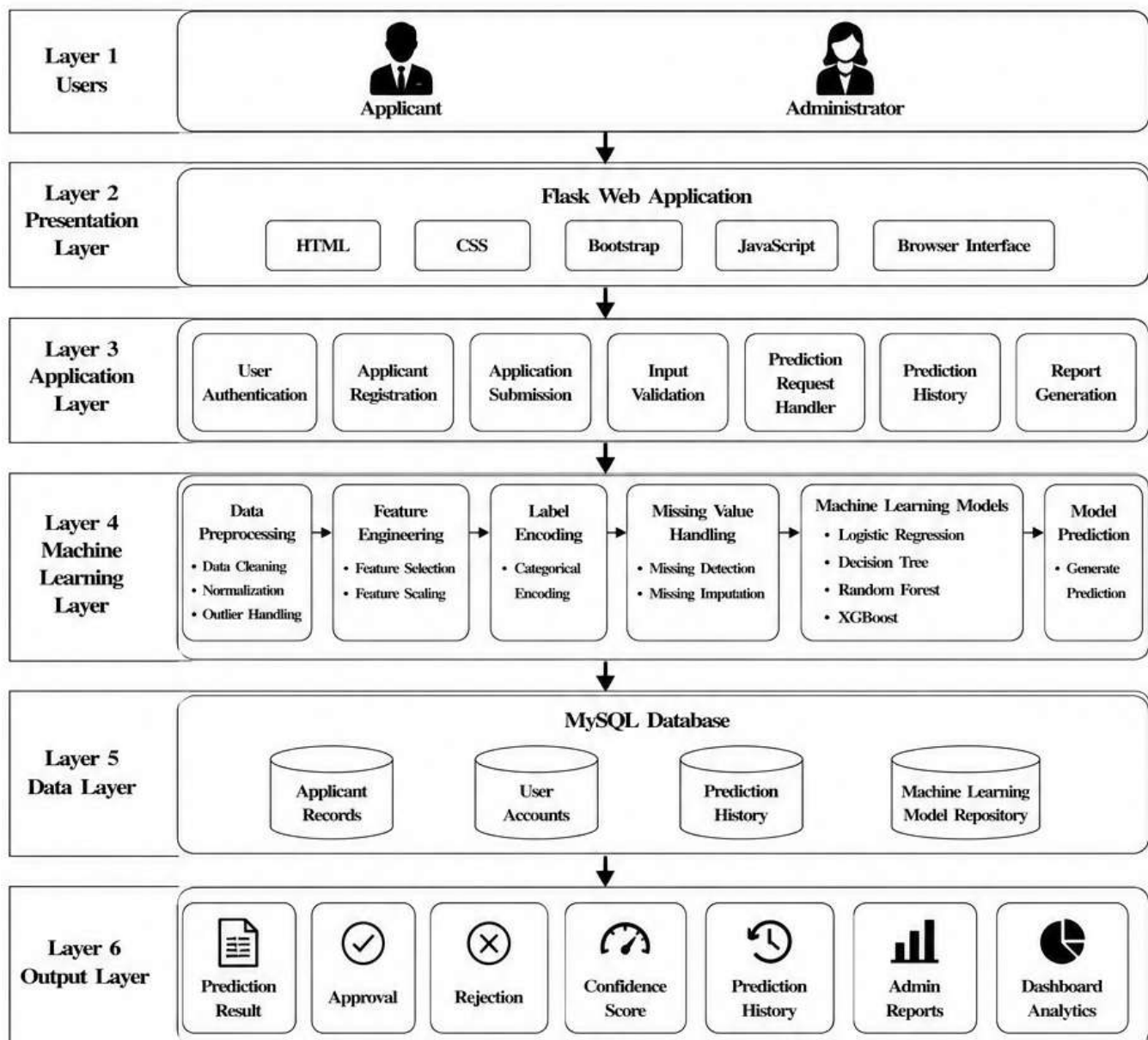


Date	03 July 2026
Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning-Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Solution Architecture

Introduction

The Solution Architecture illustrates the overall structure of the Machine Learning-Based Credit Card Approval Prediction System. It shows how users interact with the web application, how data flows through preprocessing and Machine Learning components, how prediction results are generated, and how information is securely stored within the database.



Conclusion

The layered architecture ensures modularity, scalability, maintainability, and secure communication between the web application, Machine Learning engine, and database. This design supports accurate, efficient, and reliable credit card approval prediction.

Architecture Component Description

The following table describes the responsibilities of each architectural layer and explains how they interact to provide an intelligent credit card approval prediction system.

Layer	Component	Description
Presentation Layer	Flask Web Interface	Provides responsive web pages for applicant registration, login, application submission, and prediction result display.
Application Layer	Business Logic	Handles authentication, validation, application processing, prediction requests, report generation, and system workflow.
Machine Learning Layer	Prediction Engine	Performs preprocessing, feature engineering, model loading, classification, and approval prediction using trained Machine Learning algorithms.
Data Layer	MySQL Database	Stores applicant information, user credentials, prediction history, and application records securely.
Model Repository	Serialized Machine Learning Model	Stores trained Logistic Regression, Decision Tree, Random Forest, and XGBoost models for prediction.
Security Layer	Authentication & Authorization	Protects user accounts through secure login, encrypted credentials, and role-based access control.
Output Layer	Prediction & Reports	Displays approval results, confidence scores, prediction history, dashboards, and administrative reports.

System Advantages

- Modular layered architecture
- High scalability
- Secure user authentication
- Reliable Machine Learning prediction
- Easy maintenance
- Fast response time
- Cloud deployment ready
- Supports future model updates
- Clean separation of responsibilities

Conclusion

The proposed architecture provides a secure, scalable, modular, and maintainable foundation for the Machine Learning–Based Credit Card Approval Prediction System. The layered design simplifies future enhancements while ensuring accurate and efficient prediction services.